25/02/2026

Code-A Phase-1

Aakash
Medical | IIT-JEE | Foundations
Corporate Office : AESL, 3rd Floor, Incuspaze Campus-2, Plot No. 13, Sector-18, Udyog Vihar, Gurugram, Haryana - 122015, *Ph.*+91-1244168300

MM : 720 **NCERT Booster Test Series for NEET-2026_RM(P1)_NBTS-04A** **Time : 180 Min.**

Topics Covered:
Physics: Wave Optics, Dual Nature of Radiation and Matter, Atoms, Nuclei, Semiconductor, Experimental Skills
Chemistry: The d & f-Block Elements, Coordination Compounds, The p-Block Elements, Biomolecules, Principles Related to Practical Chemistry
Botany: Molecular Basis of Inheritance, Microbes in Human Welfare, Organisms and Populations, Ecosystem, Biodiversity and Conservation
Zoology: Evolution, Human Health & Disease, Biotechnology-Principles and Processes, Biotechnology and Its Applications

General Instructions :
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

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PHYSICS

1. In Young's double slit experiment one slit has width ω while the other slit has width 4ω. Ratio of maximum intensity to minimum intensity in the interference fringe pattern is
(1) 2 : 1 (2) 4 : 1
(3) 8 : 1 (4) 9 : 1

2. In Young's double slit experiment, the 7th maxima with wavelengths λ_1 and λ_2 are at distances d_1 and d_2 from central maxima respectively. Then value of d_2 / d_1 is
(1) λ_1/λ_2 (2) λ_2/λ_1
(3) $(\lambda_1/\lambda_2)^2$ (4) $(\lambda_2/\lambda_1)^2$

3. The displacement equation of two interferring light wave are $y_1(\text{cm}) = 3\sin\omega t$ and $y_2(\text{cm}) = 4\sin\left(\omega t + \frac{\pi}{2}\right)$. The amplitude of resultant wave will be
(1) 7 cm (2) 5 cm
(3) 1 cm (4) $4\sqrt{3}$ cm

4. Shape of wavefront in case of:
I. Light diverging from a point source is spherical.
II. Light emerging out of convex lens when a point source is placed at focus is planar.
III. Parallel beam diverging from the concave lens is spherical.
Which of the above statements is/are correct?
(1) I and II only (2) II and III only
(3) I and III only (4) All I, II and III

5. Two coherent sources each emitting light of intensity I_0 interfere in a medium at a point where phase difference between them is $\frac{\pi}{3}$. The resultant intensity at that point would be
(1) I_0 (2) $3I_0$
(3) $2I_0$ (4) $4I_0$

6. A plane wavefront is incident on a concave mirror of radius of curvature (R). The radius of spherical wavefront just after reflection from the mirror is
(1) $\frac{R}{4}$ (2) $\frac{R}{2}$
(3) R (4) $2R$

7. If two coherent sources are vibrating in phase, then we have constructive interference at any point P whenever the path difference is [n is integer]
(1) $\left(n+\frac{1}{2}\right)\lambda$ (2) $\frac{n\lambda}{2}$
(3) $\left(n-\frac{1}{2}\right)\lambda$ (4) $n\lambda$

8. An unpolarised light is passing through two polaroids whose axis making an angle 60° with each other. If intensity of unpolarised light is I_0 then final intensity will be
(1) $\frac{3I_0}{8}$ (2) $\frac{I_0}{8}$
(3) $\frac{3I_0}{32}$ (4) $\frac{I_0}{32}$

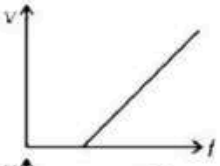
9. Match list I with list II

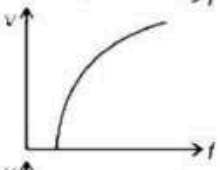
	List I		List II
a.	All bright fringes are of equal width	(i)	Diffraction
b.	Decreasing Intensity of secondary bright fringes	(ii)	Huygens principle
c.	Concept of wavefront	(iii)	Malus' law
d.	Intensity of light after passing through analyser	(iv)	YDSE

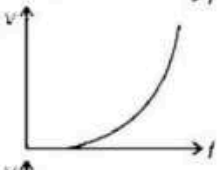
(1) a(iv), b(i), c(ii), d(iii) (2) a(i), b(ii), c(iii), d(iv)
(3) a(iv), b(ii), c(iii), d(i) (4) a(i), b(ii), c(iv), d(iii)

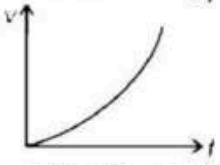
10. A narrow slit of width 2 mm is illuminated by monochromatic light of wavelength 500 nm. The distance between the first minima on either side on a screen at a distance of 1 m is
(1) 5 mm (2) 0.5 mm
(3) 1 mm (4) 0.1 mm

11. The curve drawn between the maximum speed of photoelectron (v) and frequency (f) in photoelectric effect will be

(1) 

(2) 

(3) 

(4) 

12. The de-Broglie wavelength of a neutron in thermal equilibrium with heavy water, at temperature T (Kelvin) and mass m, is
(1) $\frac{h}{\sqrt{mKT}}$ (2) $\frac{2h}{\sqrt{mKT}}$
(3) $\frac{2h}{\sqrt{3mKT}}$ (4) $\frac{h}{\sqrt{3mKT}}$

13. If the frequency of incident radiations is increased by 25%, the maximum kinetic energy of the photoelectrons emitted from photosensitive surface increases from 0.5 eV to 0.75 eV. The work function of material is
(1) 0.4 eV (2) 0.5 eV
(3) 0.6 eV (4) 0.7 eV

14. The stopping potential is directly proportional to
(1) The work function of the metal
(2) Intensity of incident radiation
(3) The saturation current for a given frequency
(4) The maximum kinetic energy gained by the photoelectrons

15. The threshold frequency for a photosensitive metal is 5×10^{13} Hz. If light of frequency 2×10^{14} Hz is incident on metal, then maximum kinetic energy of photoelectron is
(Take $h = 6 \times 10^{-34}$ J/sec)
(1) 9×10^{-20} J (2) 8×10^{-20} J
(3) 2×10^{-20} J (4) 12×10^{-20} J

16. A : Alkali metal surfaces are most suited as photo-sensitive surface.
R : The work function of alkali metals is very low.
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion.
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion.
(3) Assertion is true statement but Reason is false.
(4) Both Assertion and Reason are false statements.

17. Work function of two metal surfaces are 2.0 eV and 3.0 eV. Both are illuminated by light energy of incident photons greater than 3 eV. If graph is plotted between maximum kinetic energy and frequency for both surfaces, then slope of the graphs are in the ratio
(1) 2 : 3 (2) 3 : 2
(3) 1 : 1 (4) 4 : 9

18. If the nuclear radius of ^{64}X nucleus is 4.8 fermi then the nuclear radius of ^{125}Y nucleus is
(1) 4.8 fermi (2) 6 fermi
(3) 3.8 fermi (4) 7.5 fermi

19. If the radius of fourth orbit of hydrogen atom is r then the wavelength associated with electron particle wave in that orbit is
(1) $\frac{\pi r}{2}$ (2) πr
(3) $2r$ (4) $\frac{r}{2}$

20. When the hydrogen atom emits a photon in going from n = 5 to n = 1 stable state, then recoil speed of atom will be approximately
(1) 4 m/s (2) 8 m/s
(3) 12 m/s (4) 15 m/s

21. The force acting on the electron in a hydrogen atom depends on the principal quantum number as
(1) $F \propto n^2$ (2) $F \propto \frac{1}{n^4}$
(3) $F \propto n^4$ (4) $F \propto \frac{1}{n^2}$

22. Ratio of shortest wavelengths corresponding to Lyman and Balmer series in hydrogen spectrum is
(1) 1 : 4 (2) 5 : 27
(3) 1 : 8 (4) 3 : 4

23. Which among the following represent the Bohr radius?
(1) $\frac{h\varepsilon_0}{\pi me}$ (2) $\frac{h^2\varepsilon_0}{\pi me^2}$
(3) $\frac{h\varepsilon_0}{\pi me^2}$ (4) $\frac{h^2\varepsilon_0^2}{\pi me}$

24. An atomic hydrogen sample is excited to the nth energy level. The maximum number of spectral lines which it can emit while returning to the ground state, is
(1) $\frac{n(n+1)}{2}$ (2) $\frac{n(n-1)}{2}$
(3) $n(n-1)$ (4) $n(n+1)$

Vivek Krishna

25. The angular momentum of an electron in hydrogen atom is $\frac{5h}{2\pi}$. The kinetic energy of this electron is

(1) 10.2 eV (2) 12.09 eV
(3) 3.4 eV (4) 0.544 eV

26. The density d of nucleus varies with the mass number A as

(1) $d \propto A^3$ (2) $d \propto A^2$
(3) $d \propto A$ (4) $d \propto A^0$

27. Binding energy per nucleon is

(1) Maximum for lighter elements
(2) Maximum for heavier elements
(3) Maximum for middle order elements
(4) Equal for all elements

28. The energy liberated in the decay $^{22}_{11}Na \rightarrow {}^{22}_{10}Ne + e^+ + \nu + Q$, is

(1) $\left[m(^{22}_{11}Na) - m(^{22}_{10}Ne)\right]c^2$

(2) $\left[m(^{22}_{11}Na) - m(^{22}_{10}Ne) - 2m_e\right]c^2$

(3) $\left[m(^{22}_{11}Na) - m(^{22}_{10}Ne) - m_e\right]c^2$

(4) $\left[m(^{22}_{11}Na) - m(^{22}_{10}Ne) - 10m_e\right]c^2$

29. The binding energy of deuteron is 2.5 MeV and that of $^4_2$He is 26 MeV. If two deuterons are fused to form one $^4_2$He, then energy released in the process will be

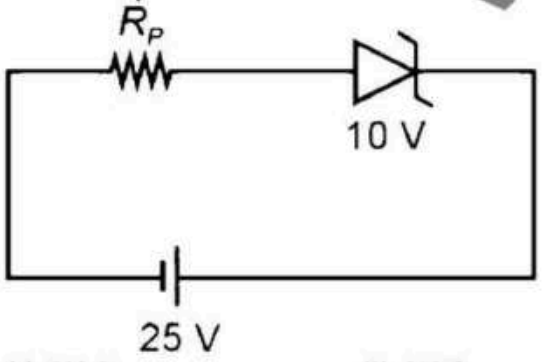
(1) 12 MeV (2) 84 MeV
(3) 21 MeV (4) 6 MeV

30. In the given nuclear reaction:

$_2He^4 + {}_ZX^A \rightarrow {}_{Z+2}Y^{A+3} + R$, R denotes

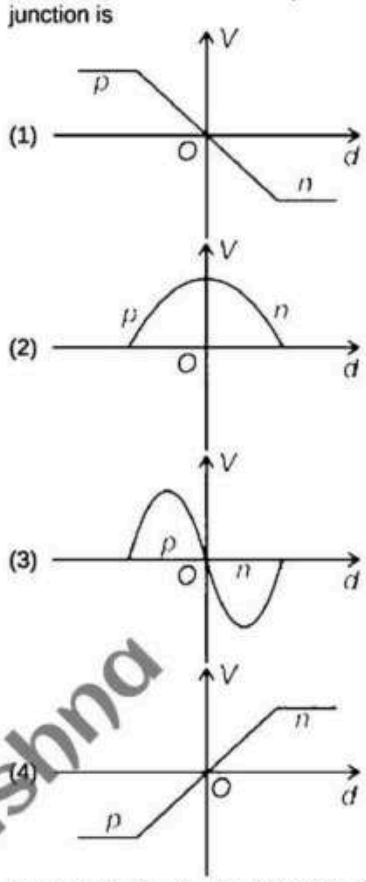
(1) Electron (2) Positron
(3) Proton (4) Neutron

31. A Zener diode having Zener voltage 10 V and power dissipation rating of 0.6 W is connected across a potential divider arranged with maximum potential drop across Zener diode as shown in the figure. The value of protective resistance R_P is

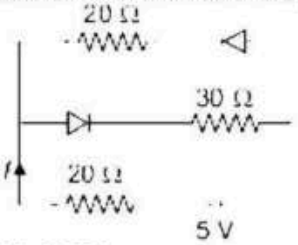


(1) 200 Ω (2) 250 Ω
(3) 300 Ω (4) 350 Ω

32. The best curve between potential and distance near p-n junction is



33. Current I in the circuit will be (All diodes are ideal)



(1) 0.05 A (2) 0.1 A
(3) 0.5 A (4) 0.2 A

34. The barrier potential of p-n junction depends on

a. Amount of doping
b. Temperature

Which of the following is correct?

(1) Only (a) (2) Only (b)
(3) Both (a) and (b) (4) Neither (a) nor (b)

35. Diffusion current in a P-N junction is less than the drift current in magnitude

(1) If the junction is forward biased
(2) If the junction is reverse biased
(3) If the junction is unbiased
(4) None of these

36. The current through an ideal p-n junction diode shown in the figure, is



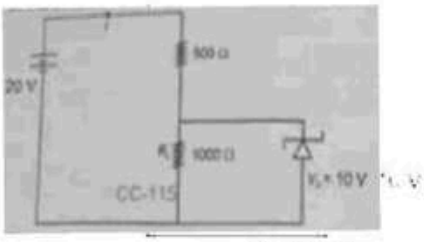
(1) 3 A (2) 3 mA
(3) 1.5 mA (4) Zero

Vivek Krishna

37. **Assertion (A) :** The depletion width decreases with increase in forward bias voltage across *p-n* junction diode.
Reason (R) : Under forward bias, the direction of the applied voltage (V) is opposite to the built-in potential V_0.
In the light of above statements, select the correct option.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true statement but (R) is false.
(4) Both (A) and (R) are false statements.

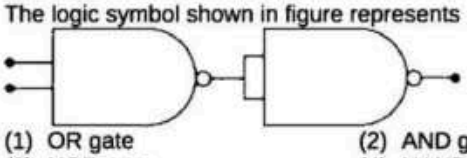
38. In half wave rectification what is the output frequency if the input frequency is 50 Hz?
(1) 50 Hz (2) 100 Hz
(3) 25 Hz (4) 75 Hz

39. In the circuit shown below, current through Zener diode is



(1) 5 mA (2) 10 mA
(3) 15 mA (4) 20 mA

40. The logic symbol shown in figure represents



(1) OR gate (2) AND gate
(3) NOR gate (4) NAND gate

41. The main scale of a vernier callipers has divisions marked in mm, 10 divisions of vernier scale coincide with 8 divisions of the main scale. The least count of the vernier callipers is
(1) 0.2 mm (2) 0.1 mm
(3) 0.5 mm (4) 0.02 mm

42. A screw gauge gives the following readings when used to measure the diameter of a wire.
Main scale reading : 2 mm
Circular scale reading : 20 divisions
Given that 1 mm on main scale corresponds to 100 divisions on the circular scale. The diameter of the wire from the above data is
(1) 0.02 cm (2) 0.022 cm
(3) 0.220 cm (4) 0.2 cm

43. The experiment which is used to determine Young's modulus of the material of a given wire, is
(1) Resonance tube experiment
(2) Displacement method
(3) Searle's experiment
(4) Young's double slit experiment

44. In a resonance tube experiment to determine the speed of sound in air, a pipe of diameter 10 cm is used. The air column in the pipe resonates with a tuning fork of frequency 425 Hz. If the minimum length of the air column is 17 cm, then the speed of sound in air at room temperature is
(1) 330 m/s (2) 350 m/s
(3) 325 m/s (4) 340 m/s

45. In a experiment to verify Ohm's law, a student measured the potential drop across the resistor as 3.90 V and current through it 1.3 A. The most suitable way to report the resistance is
(1) 3.0 Ω (2) 3.00 Ω
(3) 2.90 Ω (4) 3.1 Ω

CHEMISTRY

46. Given below are two statements:
Statement I: Iron(III) catalyses the reaction between iodide and persulphate ions.
Statement II: The correct decreasing order for first ionisation enthalpy of 3*d* series elements is:
Fe > Co > Ni > Mn > Cr
In the light of the above statements, choose the correct answer from the options given below:
(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

47. Match the processes given in List-I with their catalysts given in List-II

	List-I	List-II
(a)	Haber's process	(i) V_2O_5
(b)	Ostwald's process	(ii) Iron oxide with small amounts of K_2O and Al_2O_3
(c)	Contact process	(iii) $CuCl_2$
(d)	Deacon's process	(iv) Pt/Rh gauge

Select the **correct** option.
(1) a(i), b(ii), c(iv), d(iii) (2) a(iv), b(ii), c(i), d(iii)
(3) a(iii), b(iv), c(ii), d(i) (4) a(ii), b(iv), c(i), d(iii)

48. Which among the following ions is paramagnetic in nature?
(1) Yb^{2+} (2) Lu^{3+}
(3) Sm^{3+} (4) Ce^{4+}

49. Consider the following statements.
(a) Mixed oxides of Lanthanoids are used as catalysts in petroleum cracking.
(b) Lanthanoids are used for the production of alloy steels for plates and pipes.
(c) Mischmetall consists of approx. 95% of lanthanoid metal.
The correct statements are
(1) (a) and (b) only (2) (a) and (c) only
(3) (b) and (c) only (4) (a), (b) and (c)

50. The electronic configuration of Einstenium element is (Given atomic number of Einstenium (Es)= 99)
(1) [Rn] $5f^{10}6d^{2}7s^{1}$ (2) [Rn] $5f^{13}6d^{0}7s^{2}$
(3) [Rn] $5f^{10}6d^{1}7s^{2}$ (4) [Rn] $5f^{11}6d^{0}7s^{2}$

51. The most basic hydroxide among the following is
(1) $Tb(OH)_3$ (2) $Dy(OH)_3$
(3) $Er(OH)_3$ (4) $Pr(OH)_3$

52. Which of the following complexes exhibit maximum number of geometrical isomerism?
(1) $[Co(NH_3)_6]^{3+}$ (2) $[Pt(NH_3)_2Cl_2]$
(3) $[Co(NH_3)_2Cl_2(H_2O)_2]$ (4) $[Co(NH_3)_3Cl(CN)(H_2O)]$

Vivek Krishna

53. The correct electronic configuration of the central atom in $[Co(NH_3)_6]^{3+}$ based on crystal field theory is

(1) $t_{2g}^4 e_g^2$ (2) $t_{2g}^6 e_g^0$

(3) $e^4 t_2^2$ (4) $e^3 t_2^3$

54. Given below are two statements.

Statement I: Geometrical isomerism is not possible in tetrahedral complexes.

Statement II: $[MA_3B_3]$ type complexes show geometrical isomerism.

In the light of above statements choose the correct option given below.

(1) Both statements I and II are correct
(2) Both statements I and II are incorrect
(3) Statement I is incorrect but statement II is correct.
(4) Statement I is correct but statement II is incorrect

55. An excess of $AgNO_3$ is added to 50 mL of 0.1 M solution of tetraammineaquachloridocobalt(III) chloride. The number of mole of AgCl precipitated will be

(1) 0.1 (2) 0.02
(3) 0.03 (4) 0.01

56. Match the complex compounds in List-I with their hybridisation and shape in List-II and choose the correct option.

List-I	List-II
a. $[Fe(H_2O)_6]SO_4$	(i) sp^3, Tetrahedral
b. $[Ni(CO)_4]$	(ii) dsp^2, Square planar
c. $K_2[PtCl_4]$	(iii) d^2sp^3, Octahedral
d. $[Sc(NH_3)_6]Cl_3$	(iv) sp^3d^2, Octahedral

(1) a(iv), b(ii), c(i), d(iii) (2) a(iv), b(i), c(ii), d(iii)
(3) a(iii), b(ii), c(i), d(iv) (4) a(iii), b(i), c(ii), d(iv)

57. Which among the following is a heteroleptic complex?

(1) $K_4[Fe(CN)_6]$ (2) $K_3[Co(ox)_3]$
(3) $[PtCl_2(NH_3)_2]$ (4) $[Ni(CO)_4]$

58. Which among the following has strongest metal carbon bond?

(1) $[Mn(CO)_6]^+$ (2) $[Fe(CO)_5]$
(3) $[Co(CO)_4]^-$ (4) $[Ni(CO)_4]$

59. Which of the following pair contains both chelating ligands?

(1) H_2O and NO_2^-
(2) CN^- and NH_3
(3) NH_3 and $EDTA^{4-}$
(4) Ethane-1, 2-diamine and oxalate ion

60. IUPAC name of $K_3[Cr(C_2O_4)_3]$ is

(1) Potassium trioxalatochromate (III)
(2) Trioxalatochromate (III) potassium
(3) Potassium trioxalatochromium (III)
(4) Trioxalatochromium (III) potassium

61. Correct order of ligand field strength based on spectrochemical series is

(1) $CN^- > NH_3 > C_2O_4^{2-} > H_2O$
(2) $C_2O_4^{2-} > H_2O > NH_3 > CN^-$
(3) $C_2O_4^{2-} > NH_3 > CN^- > H_2O$
(4) $CN^- > NH_3 > H_2O > C_2O_4^{2-}$

62. Given below are two statements.

Statement I: $[NiCl_4]^{2-}$ is paramagnetic while $[Ni(CN)_4]^{2-}$ is diamagnetic.

Statement II: $[NiCl_4]^{2-}$ has 2 unpaired electrons while $[Ni(CN)_4]^{2-}$ has no unpaired electrons.

In the light of above statements choose the **correct** option.

(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

63. Consider the following complex ions

(a) $[CoF_6]^{3-}$
(b) $[Co(NH_3)_6]^{3+}$
(c) $[Fe(CN)_6]^{3-}$
(d) $[Co(H_2O)_6]^{3+}$
(e) $[MnCl_6]^{3-}$

Outer orbital complex ions are

(1) (a), (d) and (e) only (2) (a) and (e) only
(3) (b), (c) and (d) only (4) (b) and (e) only

64. Which among the following 3*d* elements has the highest enthalpy of atomisation?

(1) Ti (2) Cr
(3) V (4) Sc

65. Given below are two statements.

Statement I : β-black phosphorous is prepared by heating white phosphorous at 473 K under high pressure.

Statement II : Red phosphorous is obtained by heating white phosphorous at 573 K in an inert atmosphere.

In the light of above statements choose the correct option.

(1) Statement I is correct but statement II is incorrect
(2) Statement I is incorrect but statement II is correct
(3) Both statement I and II are correct
(4) Both statement I and II are incorrect

66. The correct order of boiling point of hydrides of group 15 elements is

(1) $NH_3 > PH_3 > AsH_3 > SbH_3$
(2) $SbH_3 > AsH_3 > PH_3 > NH_3$
(3) $SbH_3 > NH_3 > AsH_3 > PH_3$
(4) $NH_3 > SbH_3 > PH_3 > AsH_3$

67. Consider the given reaction

$$\underset{\text{(white phosphorus)}}{P_4} + NaOH + H_2O \xrightarrow{\text{Heat}} \text{'A'} + NaH_2PO_2$$

The incorrect statement about 'A' is

(1) It has a rotten fish like smell.
(2) It is a non-poisonous gas.
(3) It is weakly basic and form phosphonium compounds when reacts with acid.
(4) It reacts with mercuric chloride solution to form mercuric phosphide.

68. Consider the following reaction.

$XeF_6 + 2H_2O \rightarrow X + 4HF$

The hybridisation of central atom of X and shape of compound X respectively, are

(1) sp^3d, linear (2) sp^3d, T-shape
(3) sp^3d^2, square pyramidal (4) sp^3d, see-saw

Vivek Krishna

69. The oxoacid of sulphur that does not contain S–S linkage is
(1) $H_2S_4O_6$ (2) $H_2S_2O_7$
(3) $H_2S_2O_5$ (4) $H_2S_2O_4$

70. Consider the following statements about interstitial compounds
(i) They are chemically inert
(ii) They are very hard
(iii) They retain metallic conductivity
The correct statements are
(1) (i) and (ii) only (2) (ii) and (iii) only
(3) (i) and (iii) only (4) (i), (ii) and (iii)

71. Match List-I with List-II

	List-I (Oxides of Nitrogen formed in the reaction)		**List-II (Physical appearance and chemical nature of the oxide of nitrogen formed)**
a.	$NO + N_2O_4 \xrightarrow{250K}$	(i)	Colourless gas, neutral
b.	$Pb(NO_3)_2 \xrightarrow{673K}$	(ii)	Brown gas, acidic
c.	$HNO_3 + P_4O_{10} \rightarrow$	(iii)	Blue solid, acidic
d.	$NH_4NO_3 \xrightarrow{Heat}$	(iv)	Colourless solid, acidic

Choose the correct option
(1) a(i), b(ii), c(iii), d(iv) (2) a(iv), b(iii), c(ii), d(i)
(3) a(iii), b(ii), c(iv), d(i) (4) a(ii), b(iii), c(iv), d(i)

72. Given below are two statements
Statement I: Fluorine is a stronger oxidising agent than chlorine in aqueous medium.
Statement II: F_2 has lower bond dissociation enthalpy than Cl_2.
In the light of above statements, choose the correct option
(1) Statement I is correct but statement II is incorrect
(2) Statement I is incorrect but statement II is correct
(3) Both statement I and statement II are correct
(4) Both statement I and statement II are incorrect

73. In brown ring test, the chemical formula of brown coloured complex formed is
(1) $[Fe(H_2O)_6]^{3+}$ (2) $[Fe(H_2O)_6]^{2+}$
(3) $[Fe(H_2O)_5(NO)]^{2+}$ (4) $[Fe(CN)_5NO]^{2-}$

74. Fat insoluble vitamin among the following is
(1) Vitamin-K (2) Vitamin-C
(3) Vitamin-D (4) Vitamin-E

75. Which of the following nitrogenous base is not common in both DNA and RNA?
(1) Adenine (2) Cytosine
(3) Thymine (4) Guanine

76. Given below are two statements.
Statement I : When nucleoside is linked to phosphoric acid at 3'-position of sugar moiety we get nucleotide.
Statement II : Nucleotides are joined together by phosphodiester linkage between 5' and 3' carbon atoms of the pentose sugar.
In the light of above statements choose the correct option.
(1) Statement I is incorrect but statement II is correct
(2) Statement I is correct but statement II is incorrect
(3) Both statement I and statement II are correct
(4) Both statement I and statement II are incorrect

77. Sucrose on hydrolysis gives
(1) α-D-glucose and α-D-fructose
(2) α-D-glucose and β-D-fructose
(3) β-D-glucose and β-D-fructose
(4) β-D-glucose and α-D-fructose

78. Match List I with List II

List I (Name of vitamin)		**List II (Deficiency disease)**
a. Riboflavin	(i)	Bleeding gums
b. Thiamine	(ii)	Convulsions
c. Pyridoxine	(iii)	Digestive disorders and burning sensation of the skin
d. Ascorbic acid	(iv)	Beriberi

Choose the correct answer from the options given below.
(1) a(iii), b(iv), c(ii), d(i) (2) a(iii), b(iv), c(i), d(ii)
(3) a(iv), b(iii), c(ii), d(i) (4) a(iii), b(ii), c(iv), d(i)

79. Given below are two statements:
Statement I: Glucose does not form hydrogen sulphite addition product with $NaHSO_3$.
Statement II: α-D-(+)-glucopyranose contains four chiral centres.
In the light of above statements, choose the most appropriate answer from the options given below.
(1) Statement I is correct but statement II is incorrect
(2) Both statement I and statement II are correct
(3) Both statement I and statement II are incorrect
(4) Statement I is incorrect but statement II is correct

80. Consider the following statements
(a) Glucose does not give Schiff's test.
(b) Nitric acid oxidises gluconic acid to saccharic acid.
(c) Glucose does not reduce Fehling's solution.
The correct statements are
(1) (a) and (b) only (2) (b) and (c) only
(3) (a) and (c) only (4) (a), (b) and (c)

81. Identify the incorrect statement among the following.
(1) Starch and gum are lyophilic sols.
(2) Lyophilic sols are reversible sols.
(3) Generally, size of colloidal particle ranges from 1-1000 nm.
(4) Lyophobic sols are more stable than lyophilic sols.

82. Match the cations present in the **List-I** with their group reagent given in **List-II.**

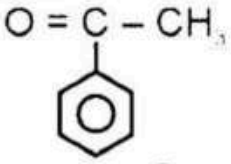
List-I	**List-II**
a. Al^{3+}	(i) H_2S in presence of NH_4OH
b. Ni^{2+}	(ii) $(NH_4)_2CO_3$ in presence of NH_4OH
c. Ba^{2+}	(iii) H_2S gas in presence of dil HCl
d. Cu^{2+}	(iv) NH_4OH in presence of NH_4Cl

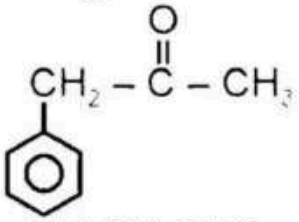
The correct match is
(1) a(i), b(iv), c(iii), d(ii) (2) a(iv), b(ii), c(i), d(iii)
(3) a(ii), b(iii), c(i), d(iv) (4) a(iv), b(i), c(ii), d(iii)

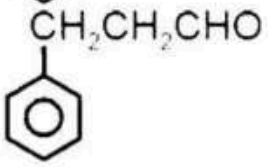
83. Which among the following is Mohr's salt?
(1) $Al_2(SO_4)_3 \cdot (NH_4)_2SO_4 \cdot 24H_2O$
(2) $K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$
(3) $FeSO_4 \cdot (NH_4)_2SO_4 \cdot 6H_2O$
(4) $K_2SO_4 \cdot Cr_2(SO_4)_3 \cdot 24H_2O$

Vivek Krishna

84. Which of the following compounds will not react with I_2/NaOH?

(1) 

(2) 

(3) 

(4) CH_3CHO

85. On addition of dilute H_2SO_4 in a salt solution, a colourless and odourless gas is evolved with brisk effervescence which turns lime water milky. Anion present in the salt may be

(1) NO_3^- (2) Cl^-

(3) NO_2^- (4) CO_3^{2-}

86. Given below are the two statements

Statement I: NH_3 gas on passing through Nessler's reagent gives a brown precipitate of basic mercury (II) amido-iodine.

Statement II: $PbSO_4$ is soluble in ammonium acetate solution due to formation of tetraacetoplumbate (II) ions.

In light of above statements, choose the **correct** answer

(1) Statement I is correct but statement II is incorrect

(2) Both statement I and statement II are correct

(3) Both statement I and statement II are incorrect

(4) Statement I is incorrect but statement II is correct

87. Working pH range of methyl orange indicator, is

(1) 3.1 to 4.5 (2) 4.2 to 6.2

(3) 6.2 to 8.2 (4) 8.2 to 10.2

88. Which of the following indicators is most suitable for weak acid vs strong base titration?

(1) Methyl orange (2) Methyl Red

(3) Phenolphthalein (4) Bromophenol blue

89. On addition of β-naphthol to benzene diazonium salt, the colour of the solution obtained is

(1) Yellow (2) Blue

(3) Scarlet red (4) Green

90. Consider the following statements:

Statement-I: Pb^{2+} ions react with potassium chromate to form yellow precipitate.

Statement-II: Lead chromate is insoluble in hot sodium hydroxide solution.

In the light of above statements, choose the correct option.

(1) Statement-I is correct and statement-II is incorrect

(2) Statement-I is incorrect but statement-II is correct

(3) Both statement-I and statement-II are correct

(4) Both the statement-I and statement-II are incorrect

BOTANY

91. Which of the following is not an example of *in-situ* conservation strategy?

(1) National park (2) Botanical garden

(3) Wildlife sanctuary (4) Sacred grove

92. The most important cause of biodiversity loss is

(1) Habitat loss and fragmentation

(2) Alien species invasion

(3) Over-exploitation

(4) Co-extinction

93. Match the Column-I with Column-II and select the **correct** answer.

Column-I	Column-II
a. Dragonflies	(i) Show species-specific insecticidal applications
b. *Glomus*	(ii) Biofertiliser
c. *Anabaena*	(iii) Forms Mycorrhiza
d. *Nucleopolyhedrovirus*	(iv) Useful to get rid of mosquitoes

(1) a(i), b(ii), c(iii), d(iv) (2) a(iv), b(iii), c(ii), d(i)

(3) a(ii), b(i), c(iii), d(iv) (4) a(iv), b(i), c(iii), d(ii)

94. Which of the following options represents the mode of action of the product produced by *Monascus purpureus*?

(1) Competitively inhibits the enzyme responsible for synthesis of cholesterol

(2) Performs fermentation to produce CO_2 to provide puffed up appearance

(3) Performs fermentation to produce alcohol

(4) Retards the growth of microbes

95. While exploring species-area relationships Alexander von Humboldt observed that

(1) Tropical environments are less seasonal

(2) There is more solar energy available in tropics

(3) There is higher productivity at tropics

(4) Within a region, species richness increased with increasing explored area, but only up to a limit

96. The bioactive molecule that is used as an immunosuppressive agent in organ-transplant patients

(1) Also acts by competitively inhibiting the enzyme responsible for cholesterol synthesis

(2) Removes clots from the blood vessels of patients

(3) Is produced by a fungus, *Trichoderma polysporum*

(4) Is produced by a bacterium, *Streptococcus*

97. Read the following statements (A-D) and select the **correct** option.

A. Species diversity decreases as we move from pole towards equator.

B. A forest in a tropical region like Equador has upto 10 times as many species of vascular plants as a forest of equal area in temperate region like the Midwest of the USA.

C. The historic convention on Biological diversity (The Earth Summit) was held in Rio de Janeiro in 2010.

D. World Summit on Sustainable Development was held in 2002 in Johannesburg, South Africa.

(1) Only A, B and D are correct

(2) Only A and D are correct

(3) Only B and D are correct

(4) Only A, B and C are correct

Vivek Krishna

98. Antibiotics means against life. This statement is in the context of
(1) Human beings
(2) Disease causing organisms
(3) All living organisms
(4) Animal beings only

99. Which among the following have maximum global diversity among plants?
(1) Angiosperms (2) Algae
(3) Mosses (4) Lichens

100. Read the following Assertion (A) and Reason (R) and chose the **correct** option
Assertion (A) : Biologists are not sure about how many prokaryotic species there might be.
Reason (R) : Conventional taxonomic methods are not suitable for identifying the species of prokaryotes in nature and many of the species of prokaryotes are non culturable under laboratory conditions.
(1) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(2) Both (A) and (R) are true and (R) is the correct explanation of (A)
(3) Both (A) and (R) are false
(4) (A) is true but (R) is false

101. Loss of biodiversity may lead to
(a) Decline in plant production.
(b) Increased resistance to drought
(c) Increased variability in plant productivity.
(1) (a) and (b) (2) (b) and (c)
(3) Only (b) (4) (a) and (c)

102. Read the following statements and select the **correct** option.
(a) Tiger census is often based on pug marks and fecal pellets.
(b) In a newly colonized habitat, emigration contributes more significantly to population growth
(c) The magnitude of intrinsic rate of natural increase for Norway rat is 0.12
(d) Co-evolution is an evolutionary mechanism where the change of biological object is triggered by the change of the related object.
(1) Only (a) and (b) are correct
(2) (a), (b) and (d) are correct
(3) Only (a) and (d) are correct
(4) (b), (c) and (d) are correct

103. Darwinian fitness can directly be represented by which of the following factors of the Verhulst-Pearl logistic growth equation given below?

$$\frac{dN}{dt} = rN\left(\frac{K-N}{K}\right)$$

(1) 'r' (2) 'N'
(3) 'K' (4) 't'

104. You never see any cattle or goats browsing on *Calotropis* because
(1) They produce highly poisonous cardiac glycosides
(2) They have spines instead of leaves
(3) They are distasteful to their herbivores
(4) They are too long in heights to reach by herbivores

105. When two closely related species competing for the same resource, cannot co-exist indefinitely and the competitively inferior one is eventually eliminated, then the interaction is known as
(1) Competition release (2) Interference competition
(3) Competitive exclusion (4) Camouflage

106. Find the **odd** one out w.r.t. predators.
(1) Predators maintain species diversity
(2) Predators are prudent in nature
(3) Bird is a predator when it consumes seeds
(4) Predators decrease prey population as they over exploit it

107. Match the given columns and select the **correct** option.

	Column I		**Column II**
a.	Aeration tank	(i)	Major part of the activated sludge is pumped into it.
b.	Primary settling tank	(ii)	Effluent undergoes constant agitation and pumping of air which allows vigorous growth of useful aerobic microbes
c.	Anaerobic sludge digesters	(iii)	Bacterial flocs are allowed to sediment
d.	Secondary settling tank	(iv)	Grit settles down to form sludge

(1) a(i), b(ii) c(iii), d(iv) (2) a(i), b(iii), c(ii), d(iv)
(3) a(ii), b(iv), c(i), d(iii) (4) a(ii), b(iv), c(iii), d(i)

108. Exponential growth
(1) Occurs under limited resources
(2) Shows a natural limit of population density for every species called carrying capacity
(3) Is influenced by environmental resistance over biotic potential
(4) Results in J-shaped growth curve

109. If the number of individuals in pre-reproductive age are less than the number of individuals in reproductive age then the population is said to be
(1) Expanding (2) Declining
(3) Stable (4) Growing

110. Epiphytes growing on mango tree is an example of
(1) Mutualism (2) Commensalism
(3) Amensalism (4) Protocooperation

111. Process of decomposition is slow if
(1) Optimum temperature and moisture is present
(2) Detritus contains more of lignin, chitin and tannins
(3) Detritus contains more amount of water soluble carbohydrates
(4) Detritus possesses more amount of nitrogenous compounds

112. How many of the following can be fermented using microbes?
a. Fish
b. Soyabean
c. Milk
d. Bamboo shoots
e. Palm sap
Choose the **correct** option from the following.
(1) Four (2) Two
(3) Five (4) Three

Vivek Krishnat

113. Pyramid of energy is always upright because
(1) Some energy is lost at each step, from particular trophic level to next trophic level
(2) A given species may occupy more than one trophic level in the same ecosystem at the same time
(3) Producers are always more in number at first trophic level in each ecosystem
(4) Energy at higher trophic level is always more than that at a lower trophic level

114. 100 *E. coli* cells with ^{14}N dsDNA are incubated in medium containing ^{15}N nucleotides. After 40 minutes, how many *E. coli* cells will have DNA totally free from ^{14}N?
(1) 200 cells (2) 80 cells
(3) 100 cells (4) 400 cells

115. Grasses → Grasshopper → Birds → Hawk.
What percent of energy is available to birds if at T_4 level energy is 40 J?
(1) 4 kJ (2) 0.4 kJ
(3) 4 J (4) 400 kJ

116. Pyramid of biomass in terrestrial ecosystem is
(1) Never upright (2) Spindle shaped
(3) Upright (4) Always inverted

117. Read the following statements and select the **correct** option.
Statement A : Gross primary productivity of an ecosystem is the rate of production of organic matter during photosynthesis.
Statement B : Net primary productivity is the available biomass for the consumption to herbivores and decomposers.
(1) Only statement **A** is correct
(2) Only statement **B** is correct
(3) Both the statements **A** and **B** are correct
(4) Both the statements **A** and **B** are incorrect

118. All of the following are limitations of ecological pyramids **except**
(1) It does not take into account the same species belonging to two or more trophic levels.
(2) It assumes a simple food chain
(3) Saprophytes are not given any place in ecological pyramids
(4) Biomass and energy level of organism can be depicted in it

119. Read the following assertion (A) and reason (R) and choose the **correct** option.
Assertion (A) : In a natural ecosystem some animals like cockroaches, crows etc. are omnivores.
Reason (R) : When any organism dies, it is converted to detritus, which serves as an energy source for decomposers.
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

120. In competition, herbivores are more adversely affected than carnivores because
(1) Former have less alternative resources for their survival as compared to carnivores
(2) Resources are never limiting for herbivores
(3) Herbivores are always fewer in number than carnivores
(4) Latter being at higher trophic levels have high number of predators

121. Standing crop refers to the
(1) Total primary production of crop field in a period of time
(2) Mass of living material in an area at a particular time
(3) Amount of inorganic substances in a detritus food chain
(4) Total amount of transferable energy from producers to primary consumers in a grazing food chain

122. The distance between two consecutive base pairs according to Watson and Crick model of DNA is
(1) 0.34 mm (2) 0.34 nm
(3) 3.4 nm (4) 3.4 mm

123. Read the given statements and choose the **correct** option.
Statement A : The untranslated regions on mRNA are required for efficient translation process.
Statement B : Splicing is the process of removal of introns and joining of exons in a defined order.
(1) Only statement A is correct
(2) Both statement A and statement B are incorrect
(3) Only statement B is correct
(4) Both statement A and statement B are correct

124. Match the Column I with Column II and select the **correct** option.

Column I	Column II
a. Frederick Griffith	(i) Preparation of polynucleotide using synthetic DNA
b. Avery, MacLeod and McCarty	(ii) Transforming principle
c. Har Gobind Khorana	(iii) Semiconservative replication of DNA
d. Meselson and Stahl	(iv) Identifying biochemical nature of transforming principle

(1) a(iv), b(ii), c(i), d(iii) (2) a(ii), b(i), c(iii), d(iv)
(3) a(ii), b(iv), c(i), d(iii) (4) a(iii), b(i), c(iv), d(ii)

125. The number of glycosidic bonds associated with dsDNA of λ-phage is
(1) 48500 (2) 48502
(3) 97004 (4) 97002

126. If a dsDNA contains 20% of A, what will be the percentage composition of G?
(1) 20% (2) 40%
(3) 30% (4) 60%

127. Feature(s) providing stability to the helical structure of DNA is/are
(a) Presence of histone proteins
(b) H-bonds between the two strands
(c) Plane of one base pair stacking over the other
(d) Presence of Adenine
(1) (a) and (d) (2) (d) only
(3) (b) only (4) (b) and (c)

128. Select the **correct** match w.r.t. population interaction.

Name of Interaction	Species A	Species B
(1) Parasitism	+	+
(2) Mutualism	+	–
(3) Predation	+	+
(4) Commensalism	+	0

(1) (1) (2) (2)
(3) (3) (4) (4)

129. During the process of transcription RNA polymerase II transcribes
(1) snRNA and tRNA (2) hnRNA
(3) rRNA (4) 5S rRNA

Vivek Krishna

130. Select the **incorrect** statement w.r.t. polymorphism.
(1) Polymorphism arises due to mutations
(2) Polymorphism in DNA sequence is the basis of DNA fingerprinting
(3) Polymorphism represents variation at genetic level
(4) Polymorphism do not play any role in evolution and speciation

131. DNA fingerprinting technique was initially developed by
(1) Marshall Nirenberg (2) Alec Jeffreys
(3) Francois Jacob (4) Alfred Hershey

132. In human genome, SNPs are
(1) The sequence of whole set of genes
(2) Groups of all the junk DNA
(3) The locations where single base DNA differences occurs
(4) Stretches of DNA sequences that are repeated many times

133. Opening of DNA helix during transcription is performed by
(1) Helicase (2) RNA polymerase
(3) Ligase (4) Topoisomerase

134. 5'UAACCAUAGCU3' is a sequence of newly synthesized primary transcript. Select the **correct** option for the nucleotides sequence in coding strand of DNA.
(1) 5'AUUGGUAUCGA3' (2) 5'AGCUAUGGUUA3'
(3) 3'ATTGGTATCGA5' (4) 5'TAACCATAGCT3'

135. Match the list I with list II w.r.t. *lac* operon and choose the correct option

	List I		List II
a.	*z* gene	(i)	Associated to increase the permeability of the cell to beta galactosides
b.	*y* gene	(ii)	It codes for the enzyme responsible for the hydrolysis of a diasaccharide
c.	*i* gene	(iii)	It encodes for transacetylase
d.	*a* gene	(iv)	Constitutive gene

(1) a(ii), b(i), c(iv), d(iii) (2) a(iii), b(iv), c(i), d(ii)
(3) a(ii), b(iv), c(iii), d(i) (4) a(iii), b(ii), c(iv), d(i)

ZOOLOGY

136. Consider the following features
(a) Lived in East African grasslands 2 mya
(b) They were the first human like being, the hominid.
(c) Brain capacity was more than 650-800 cc
(d) Hunted with stone weapons but essentially ate fruits
Which of the above features are true for *Australopithecines*?
(1) (a) and (d) (2) (b) and (c)
(3) (a), (b) and (d) (4) (b), (c) and (d)

137. Which amongst the following is the most recent ancestor of the existing mammals?
(1) Thecodonts (2) Sauropsids
(3) Synapsids (4) Therapsids

138. Horsetails which were abundant in carboniferous period were descendants of
(1) Zosterophyllum (2) Ferns
(3) Psilophyton (4) Lycopods

139. Match column I with column II.

	Column-I		Column-II
a.	First cellular form of life appeared on Earth	(i)	About 500 mya
b.	Jawless fish probably evolved	(ii)	About 65 mya
c.	Dinosaurs suddenly disappeared	(iii)	About 350 mya
d.	Invertebrates formed and active	(iv)	About 2000 mya

Select the **correct** option.
(1) a(iii), b(iv), c(ii), d(i) (2) a(ii), b(i), c(iii), d(iv)
(3) a(i), b(ii), c(iii), d(iv) (4) a(iv), b(iii), c(ii), d(i)

140. Given below are two statements, one is the assertion (A) and the other is reason (R).
Assertion (A): When we describe the story of life on Earth, we treat evolution as a consequence of a process called natural selection.
Reason (R) : Natural selection is not at all based on factual observations and is independent of availability of natural resources or seasonal fluctuations.
In the light of above statements, select the most appropriate option.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

141. Read the given statements.
Statement A: Few fossils of man-like bones discovered in Ethiopia and Tanzania revealed hominid features, indicating that man-like primates walked in eastern Africa.
Statement B: Continental drift caused North America to join South America, overriding North American mammals by South American fauna.
Choose the **correct** option.
(1) Both statements A and B are correct
(2) Both statements A and B are incorrect
(3) Only statement A is incorrect
(4) Only statement B is incorrect

142. Choose the **correct** match.
(1) Hugo deVries – Theory of inheritance of acquired characters
(2) Oparin and Haldane – Theory of chemical evolution
(3) Lamarck – Mutation theory
(4) Louis Pasteur – Supported the theory of spontaneous generation
(1) (1) (2) (2)
(3) (3) (4) (4)

143. S. L. Miller observed the formation of which of the following during his electric discharge experiment?
(1) Pigments (2) Lipids
(3) Carbohydrates (4) Amino acids

Vivek Krishna

144. When migration of a section of population happens multiple times to another place and population, then this phenomenon would be called
(1) Gene migration (2) Gene flow
(3) Genetic drift (4) Genetic equilibrium

145. All of the following indicates common ancestry, **except**
(1) Wings of butterfly and that of birds
(2) Forelimbs of cheetah and bat
(3) Brains of frog and that of human
(4) Thorn of *Bougainvillea* and tendrils of *Cucurbita*

146. Earth was supposed to have been formed about
(1) 500 million years ago (2) 4.5 billion years back
(3) 2000 million years ago (4) 450 million years ago

147. All of the following plants produce certain substances which cause hallucinations at certain doses, **except**
(1) *Erythroxylum coca* (2) *Atropa belladona*
(3) *Datura* (4) *Papaver somniferum*

148. Select the **incorrect** statement w.r.t antibody.
(1) Antigen binding site is present on N-terminal amino acids of heavy and light chains.
(2) Antibodies are found in the blood.
(3) Each monomeric antibody molecule contains two types of peptide chains.
(4) Overall, there are 12 disulfide bonds present in a monomeric antibody molecule.

149. Select the **correct** option w.r.t AIDS.
(1) Macrophages act like HIV-factory.
(2) Macrophages cannot survive while viruses are being replicated and released.
(3) Widal test is the widely used diagnostic test.
(4) Treatment with anti-retroviral drugs are fully effective and can cure the disease.

150. How many of the organs mentioned in the box below are involved in differentiation of immature lymphocytes into antigen-sensitive lymphocytes in humans?

Bone marrow, Spleen, Tonsils, Thymus, Lymph nodes

Choose the **correct** option.
(1) Three (2) Four
(3) Two (4) Five

151. Match column I and column II w.r.t diseases and vectors/carriers of their pathogens.

Column I	**Column II**
a. Malaria	(i) *Culex*
b. Dengue	(ii) House fly
c. Amoebiasis	(iii) *Anopheles*
d. Filariasis	(iv) *Aedes*

Select the **correct** option.
(1) a(i), b(ii), c(iv), d(iii) (2) a(ii), b(iii), c(iv), d(i)
(3) a(iii), b(iv), c(ii), d(i) (4) a(i), b(iii), c(iv), d(ii)

152. The lobed organ, located near the heart and beneath the breastbone, which is quite large at the time of birth but keeps reducing in size with age, is
(1) Bone marrow (2) Thymus
(3) Spleen (4) Appendix

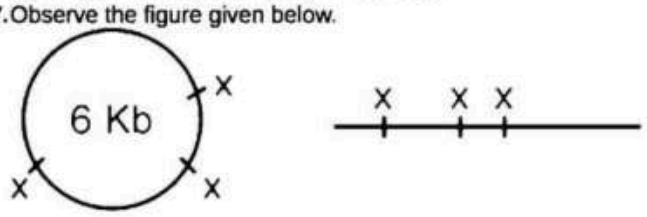
153. Read the given statements.
Statement I : Health, for a long time by early Greeks, was considered as a state of body and mind where there was a balance of certain 'humors'.
Statement II : Both neural system and endocrine system help in maintaining our health by influencing immune system.
Select the **correct** option.
(1) Only statement I is correct
(2) Only statement II is correct
(3) Both statements I and II are correct
(4) Neither statement I nor statement II is correct

154. Sustained high fever [39 °C to 40°C], weakness, stomach pain, constipation, headache and loss of appetite are the common symptoms of a bacterial disease which can be confirmed by
(1) Widal test (2) Biopsy
(3) Bone marrow test (4) Computed tomography

155. Computed tomography uses 'A' to generate a three-dimensional image of the internals of an object. MRI uses 'B' to accurately detect pathological and physiological changes in the living tissue.
Identify 'A' and 'B' respectively and select the correct option.
(1) UV-rays, X-rays
(2) X-rays, non-ionising radiations
(3) UV-rays, γ-rays
(4) Non-ionising radiations, ionising radiations

156. Choose the correct set of diseases that spread through contaminated food and water.
(1) Amoebic dysentery, Ascariasis, Polio
(2) Plague, Filariasis, Diphtheria
(3) Tetanus, Dengue, Malaria
(4) Elephantiasis, Small pox, Amoebiasis

157. Observe the figure given below.



In the above circular DNA (A) and linear DNA (B), there are three restriction sites for restriction enzyme X which are not equidistant to each other.
On complete digestion of (A) and (B), the numbers of respective DNA bands visible on electrophoretic plate will be
(1) Three, two (2) One, three
(3) Three, four (4) Four, three

158. All of the following are steps included under downstream processing, **except**
(1) Separation of protein (2) Purification of protein
(3) Preservation of product (4) Expression of gene

159. If an alien piece of DNA is inserted in restriction site of *Pst* I in pBR322, then the resultant recombinants will be
(1) Ampicillin resistant
(2) Tetracycline and ampicillin resistant
(3) Tetracycline susceptible
(4) Ampicillin susceptible

Vivek Krishna

160. Which of the following organisms is **not** correctly matched with its cell wall degrading enzyme?
(1) Plant cells – Cellulase (2) Green algae – Methylase
(3) Fungi – Chitinase (4) Bacteria – Lysozyme

161. During isolation of genetic material, the purified DNA ultimately precipitates out after the addition of
(1) Chilled ethanol (2) Chitinase
(3) Ribonuclease (4) DNA ligase

162. Which of the following can assist in identifying the transformed cells but is absent in pBR322?
(1) *rop* gene (2) β-galactosidase
(3) *Bam*HI (4) *Pvull*

163. Arrange the following steps involved in rDNA technology in correct order.
(a) Isolation and separation of the DNA fragment.
(b) Cutting of DNA at specific locations.
(c) Isolation of the genetic material.
(d) Ligation of DNA fragment into vector.
Choose the **correct** option.
(1) a → c → b → d (2) c → b → a → d
(3) b → a → c → d (4) d → a → b → c

164. A method of gene transfer which involves bombardment of high velocity microparticles of gold coated with DNA is mainly used for
(1) *Meloidogyne* (2) *Rattus*
(3) Plants (4) Nematodes

165. An enzyme belonging to _____ plays a vital role in the formation of rDNA as it helps in linking the alien DNA and cloning vector and is also called _____.
Select the correct option to fill in the blanks respectively.
(1) Class VI; molecular scissors
(2) Class IV; molecular glue
(3) Class VI; molecular glue
(4) Class IV; molecular scissors

166. If the enzyme *Taq* polymerase fails to function in PCR, then which of the following would most likely to get affected?
(1) Denaturation
(2) Annealing
(3) Extension
(4) Separation of template DNA strands

167. Which of the following scientists developed a method of utilising small ringlets of DNA as the vector?
(1) Herbert Boyer (2) Stanley Cohen
(3) S. L. Miller (4) J.D. Watson

168. All of the statements given below are correct w.r.t. gel electrophoresis, **except**
(1) Smallest DNA fragment is present near anode.
(2) DNA fragments separate due to sieving effect provided by gel.
(3) Separated DNA fragments are extracted from the gel piece through a process called spooling.
(4) DNA fragments move towards the positive electrode due to their negative charge.

169. How many of the following mentioned in the box below is/are genetically modified?
Pomato, Retrovirus, *Meloidogyne incognita*, Rosie, *Bacillus thuringiensis*
Choose the **correct** option.
(1) Five (2) Four
(3) Three (4) One

170. In 1990s, an American company got patent rights on a crop derived from Indian semi-dwarf varieties through US Patent and Trademark Office by claiming it as an invention. Similar attempts have been made to patent the use of product of traditional herbal medicine such as
(1) Wheat (2) Tobacco
(3) Neem (4) Bt soyabean

171. Today transgenic models exist for human diseases such as
(a) Cancer
(b) Alzheimer's disease
(c) Cystic fibrosis
Choose the correct option.
(1) Only (b) and (c) (2) Only (a)
(3) (a), (b) and (c) (4) Only (c)

172. Choose the **correct** statement.
(1) Insulin is synthesised in mature form by joining three peptide chains namely chain A, B and C.
(2) Earlier insulin was extracted from gall bladder of the live cattle and pigs.
(3) In 1988, an American company prepared three polypeptides corresponding to namely chain A, B and C chains and introduced them in plasmids.
(4) Chains A and B were produced separately in *E.coli*, extracted and combined by creating two interchain disulfide bonds to form recombinant human insulin.

173. To introduce genetically modified cow that produce protein enriched milk in Indian markets, permission from which of the following organisations will be necessary?
(1) NACO (2) GEAC
(3) WHO (4) EFB

174. Select the **incorrect** statement w.r.t. transgenic animals.
(1) 95 per cent of all the existing transgenic animals are mice.
(2) Transgenic monkeys are being developed for safety testing of vaccines and can replace mice very soon.
(3) They are made to carry genes which make them more susceptible to toxic substances.
(4) They are utilised to form biological products for the treatment of diseases like phenylketonuria.

175. What could be the permanent cure for a genetic disease caused due to the deficiency of adenosine deaminase?
(1) Gene therapy after birth
(2) Gene therapy during early embryonic stage
(3) Bone marrow transplantation at later stage of life
(4) Enzyme replacement therapy

176. The plants produced by micro-propagation are genetically identical to the original plant from which they are grown and are called
(1) Explants (2) Meristems
(3) Somaclones (4) Somatic hybrids

Vivek Krishna

177. **Assertion (A):** After industrialisation in England, the population of white-winged moth declined significantly while melanised moths flourish.
Reason (R): During post-industrialisation period, the tree trunk became dark due to deposition of industrial smoke and soot that helped predators to spot a white-winged moth against a contrasting background and not the dark-winged moths.
In the light of above statements, choose the correct answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true, (R) is false
(4) Both (A) and (R) false

178. A novel mechanism of RNA interference inhibits
(1) Transcription of specific mRNA of the host plant
(2) DNA replication within the host plant
(3) Translation of specific genes in prokaryotes
(4) Translation of specific mRNA in eukaryotes

179. Presently ____ recombinant therapeutics have been approved for human-use world over and _____ of these are being marketed for human-use in India.
Select the **correct** option to fill in the blanks respectively.
(1) 40; 20
(2) 30; 12
(3) 12; 30
(4) 35; 15

180. Which of the following is a modern diagnostic technique where an infection can be detected by the presence of glycoproteins of pathogen or by detecting immunoglobulin formed against the pathogen?
(1) MRI
(2) Urine analysis
(3) ELISA
(4) Serum analysis

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